

Reproducible model-based AI planning for county-scale farmland consolidation in fragmented mountain landscapes

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Abstract

Farmland consolidation is a spatial-planning lever for mechanisation, slope-erosion control, and large-patch formation in mountainous counties, yet county-scale optimisation has remained out of reach for routine planning offices: the cadastral action space is high-branching, rewards are multi-objective, and reinforcement-learning baselines on this task require 8–12 GPU-hours per seed. Here we show that a learned transition-model and model-predictive-control pipeline, trained in 25 min and planning each scenario in ~ 5 min on a desktop CPU, reduces area-weighted farmland slope by -2.001% on Bishan District (independent GIS-only verification on 53,004 swappable parcels), shifting 687 ha out of the steep ($\geq 15^\circ$) bands that GB/T 21010-2017 flags as marginal for cultivation; the result reproduces on Neijiang Dongxing (76,376 parcels). On the matched research-side cadastre on which we trained in-house centralised-PPO and multi-agent reinforcement-learning baselines, the lab pipeline reaches $-1.289 \pm 0.079\%$ across five independently trained ensembles, $1.6\times$ the magnitude reached by either DRL baseline; against simulated annealing under a matched 180 s wall budget the reward gap is $9.6\times$. An auxiliary pairwise ranking loss resolves a hidden failure of the mean-squared-error objective when within-state reward variance is small relative to between-state variance, the regime that farmland inhabits. A public-data US restoration cross-domain case and a simulator-cost crossover analysis bound the regime where the pipeline beats classical operations-research baselines. The full pipeline ships as an ArcGIS Pro toolbox and a non-commercial CLI.

1 Introduction

Mountainous and hilly landscapes contribute a substantial share of global agricultural production but pose management challenges distinct from contiguous large-field plains: small, irregular cadastral parcels distributed across steep terrain raise per-hectare labour cost, prevent the use of medium-scale machinery, increase vulnerability to climate-driven yield shocks, and limit policy levers for raising productivity without displacing farmers [Romeo et al., 2020, Liu and Li, 2023, Niu et al., 2023, Foley et al., 2011, Springmann et al., 2018]. Land consolidation—the spatial rearrangement of farmland and forest parcels within legal and topographic constraints—is the dominant policy response [Liu et al., 2014, Long, 2014, Rounsevell et al., 2012]. In China, where a substantial fraction of cultivated land lies in hilly or mountainous counties [Huang et al., 2018], consolidation is operationalised through the formation of *baimu fang* (literally “hundred-*mu* patches”, ≥ 6.67 ha of contiguous farmland) and through township-level regulatory plans that bound feasible swaps between farmland and forest classes.

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Computationally, county-scale consolidation is a high-dimensional, multi-objective combinatorial planning problem: a single county routinely contains tens of thousands of parcels, three competing objectives (slope reduction for mechanisation, contiguity for shape, large-patch consolidation for management), and hard adjacency and tenure constraints. Classical approaches—rule-based heuristics, genetic algorithms [Cao et al., 2012, Deb et al., 2002, Stewart et al., 2004, Aerts and Heuvelink, 2002], and model-free deep reinforcement learning [Zheng et al., 2023, Equihua et al., 2024, Janowicz et al., 2020]—all face structural limits at this scale: heuristics ignore the multi-objective trade-off, metaheuristics miss long-horizon dependencies, and model-free RL in our experience requires 8–12 GPU-hours per run and fails on 25–40% of random seeds in the discrete county-scale regime, costs that prevent iterative scenario exploration by planning offices.

Model-based reinforcement learning offers a structurally appealing alternative: learn a transition model once from data, then exploit it for cheap planning [Chua et al., 2018, Janner et al., 2019]. Recent work in continuous control [Hafner et al., 2025, Hansen et al., 2024] and game-playing [Schrittwieser et al., 2020, Silver et al., 2017] demonstrates the paradigm; AI-for-sustainability and Earth-observation work has begun to adopt these methods at policy scale [Rolnick et al., 2022, Reichstein et al., 2019, Tuia et al., 2024]. We use *learned planning model* for a feed-forward deterministic next-state predictor with a discriminative reward head, rather than the recurrent latent “world model” of the Dreamer family, because our setting is fully observed and episodic on pre-extracted cadastral block features.

Contribution. This manuscript reports three contributions, in decreasing order of generality. First, we present an *empirical characterisation* of when learned-surrogate model-predictive control delivers an operationally meaningful advantage over both reinforcement-learning and classical operations-research baselines on county-scale spatial planning, anchored in two real Chinese counties (Bishan, Chongqing, 52,515 parcels; Neijiang Dongxing, Sichuan, 76,376 parcels) plus a public-data US restoration case (Buchanan County, Virginia, 562 units). Second, we package this characterisation as a *falsifiable empirical diagnostic* that a practitioner can evaluate before adopting the pipeline: the within-state-to-between-state reward variance ratio σ_a/σ_s predicts whether an auxiliary pairwise ranking loss is required to make standard mean-squared-error (MSE) training planning-useful (small ratios near 0.6, as on Bishan, require it; ratios in the 5–7 range, as on Buchanan, do not), and the simulator step cost—17.9 ms on the farmland environment vs 0.05 ms on the restoration environment in our measurements—combined with whether the reward exhibits combinatorial cost interactions, predicts whether a learned surrogate is worth its training overhead at all. We frame both as empirical diagnostics calibrated on the present evidence rather than universal thresholds. Third, we release a *fully reproducible deployment pipeline*: trained ensembles, a deterministic seven-preset synthetic benchmark under CC-BY 4.0, the public-data Buchanan restoration case, an ArcGIS Pro Python Toolbox, a non-commercial CLI workflow, and a ~ 70 -second smoke test. On the headline Bishan deployment, an independent GIS-only recomputation of the on-disk optimised cadastre reports a slope reduction of -2.001% relative to the initial 9.62° (Methods §9), with 687 ha shifted out of the $\geq 15^\circ$ steep bands and a contiguity gain of $+0.013$. On the research-side cadastre that matches the pipeline our in-house centralised-PPO and multi-agent RL baselines were trained on, five independently trained contrastive ensembles reach $-1.289 \pm 0.079\%$, $1.6\times$ the reduction reached by either DRL baseline (§4, Welch’s t -test, $|d| > 1.9$), while reducing wall time by roughly two orders of magnitude; the cross-county Neijiang result reproduces the qualitative finding. The auxiliary pairwise large-margin loss [Burges et al., 2005, Joachims, 2002] that operationalises the first half of the diagnostic (raising within-state ranking accuracy from 51.6% to 85.5% and converting model-predictive control from worst variant tested to best) is the algorithmic mechanism

behind these numbers, not the contribution’s headline; pairwise-preference training has been used in surrogate-assisted optimisation and learning-to-rank for two decades [Joachims, 2002, Burges et al., 2005, Christiano et al., 2017, Ouyang et al., 2022] and the present novelty is its empirical role and boundary in spatial planning rather than its derivation as a learning rule.

2 Multi-objective consolidation as a planning problem at county scale

We anchor the analysis in two complete county-scale datasets from southwest China: Bishan District in Chongqing (13 townships, 52,515 parcels grouped into 2,600 spatial blocks) and Neijiang Dongxing District in Sichuan (29 townships, 76,376 parcels grouped into 3,711 blocks; per-county initial-state metrics are reported in Supplementary Table S1). Together the two counties comprise 128,891 individual cadastral parcels, an order of magnitude larger than published consolidation benchmarks we are aware of [Cao et al., 2012, Zheng et al., 2023]. Area-weighted mean farmland slope is 9.62° on Bishan and 10.55° on Neijiang—both above the 6° terracing threshold of the Chinese High-Standard Farmland Construction standard (GB/T 30600-2022) beyond which mechanised cultivation requires terraced fields. The within-farmland contiguity index is 3.59 on Bishan and 2.63 on Neijiang. The legally relevant consolidation unit, the *baimu fang* (≥ 6.67 ha contiguous farmland eligible for mechanised management), is unevenly distributed: 109 patches totalling 46,844 ha on Bishan, 384 patches totalling 74,342 ha on Neijiang.

Three properties make the problem hard. The action space is combinatorial—across a 100-step episode the joint decision space is approximately 2600^{100} on Bishan alone, vastly larger than any standard combinatorial benchmark in the spatial-planning literature. The objectives interact: slope-reducing swaps near fragment boundaries often retire marginal farmland from the largest existing patches, so a planner who optimises slope greedily can lose qualifying area, and vice versa. The wall-time profile of existing methods rules out iterative scenario exploration: in-house Centralised PPO and multi-agent baselines on the same task require 8–12 GPU-hours of training per seed and Centralised PPO collapses on 2/5 seeds (Supplementary Section S1). A successful plan must reduce farmland slope, raise contiguity, and increase the qualifying *baimu fang* count simultaneously, under hard paired farmland–forest swap constraints and a township-level swap budget that throttles the rate at which any one township sheds cultivated area; the cultivated-land red line is enforced by an explicit constraint at deployment time (Methods, §9).

3 Diagnosing and fixing a hidden ranking failure in the standard learned model

We trained a learned planning model—an ensemble of three transition networks ($\sim 237,000$ parameters each, mean-squared-error objective)—on 6,000 transitions sampled from random and greedy policies in the Bishan environment. By conventional fidelity metrics the model is excellent: state cosine similarity exceeds 0.998, training and validation losses converge cleanly, and the model is rolled forward autoregressively for 25 steps before predictions diverge perceptibly (*s*-cosine drops from 0.999 at one step to 0.94 at 50 steps).

The aggregate metrics, however, conceal a near-total failure of per-action discrimination. We constructed a pairwise-action benchmark by snapshotting the simulator state at each of 1,000 sampled environment states, executing 50 valid actions one at a time, recording the immediate ground-truth reward, and restoring the snapshot before continuing. The MSE-trained ensemble

correctly ranks only 51.6% of action pairs by predicted reward—statistically indistinguishable from random—while predicting a reward standard deviation across the action menu of 0.274 versus the ground-truth 0.811, a three-fold compression of the discriminative signal. A fresh re-train with the identical objective reaches 60.2% ranking accuracy on the same benchmark; both points sit well below the 85.5% that the auxiliary ranking loss subsequently delivers (Table 1). We therefore treat 51.6% as the lower envelope of MSE-only behaviour at this σ_a/σ_s regime rather than its central tendency, and the 25–34 pp gap between MSE-only and contrastive ensembles—rather than the absolute level reached by any single MSE-only retrain—as the empirical fingerprint of the diagnostic. Four independent operational failures follow at the deployment layer: dream-trained policies collapse under uncertainty-guided evidence collection because 99.6% of candidate transitions are filtered out by an ensemble-disagreement signal that carries no actual signal (the standard ensemble disagreement diagnostic shows Pearson $r = -0.024$ between disagreement and absolute prediction error); the random-sampling MPC ceiling is $-0.209 \pm 0.008\%$; every aggressive-exploitation variant fails catastrophically (greedy continuation gives $+0.029\%$, the wrong sign, and slope-only scoring gives -0.001%); and doubling the candidate budget to $K=100$ provides no improvement.

The decomposition behind this is structural to the loss. In plain terms, two quantities govern whether MSE training delivers a planning-useful reward predictor. The between-state spread σ_s measures how much average reward varies from one state to the next; the within-state spread $\sigma_a(s)$ measures how much reward varies between the candidate actions available at a single state. Planning depends almost entirely on the latter—top- K selection compares actions inside a single state, never across states—but MSE training is dominated by the former whenever $\sigma_a(s)$ is small relative to σ_s , because the loss-minimising predictor in that regime is the conditional mean $\bar{r}(s)$ which assigns the same reward to every action regardless of identity. On Bishan, the empirical ratio is $\sigma_a/\sigma_s \approx 0.6$, the regime where the conditional-mean predictor gives random ranking by construction. Aggregate prediction fidelity (cosine 0.998) and within-state ranking accuracy (51.6%) are therefore not constrained against each other under MSE training, and the latter is what model-predictive top- K selection actually consumes; we give the formal statement and its evaluation-methodology implications in Methods (§9).

The diagnosis points at a concrete intervention: add an explicit pairwise large-margin ranking term [Herbrich et al., 2000, Joachims, 2002, Burges et al., 2005, Burges, 2010] to the model’s reward head, applied alongside the existing reconstruction objective (Methods, §9):

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{state}} + \mathcal{L}_{\text{global}} + 0.1 \mathcal{L}_{\text{reward}}}_{\text{MSE terms}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}. \quad (1)$$

The signal is applied to the existing reward head, not a separate reward proxy, so deployment requires no additional model. This is a model-side use of pairwise preferences, distinct from the policy-side RLHF formulation [Christiano et al., 2017, Ouyang et al., 2022]. Training data are inexpensive: 1,000 states sampled by random rollout, 50 valid actions snapshotted/executed/restored at each, generated in ~ 45 min on commodity CPU.

The intervention closes the discriminative gap with no measurable cost to aggregate fidelity (Table 1). Pairwise ranking accuracy rises from 60.2% ($\lambda_{\text{rank}}=0$) to 85.5% ($\lambda_{\text{rank}}=5.0$) while state-prediction cosine similarity stays above 0.999. The qualitative effect on MPC is a reversal: greedy continuation, the variant most cleanly attributable to ranking error, becomes the best variant, and cross-seed mean slope improvement on Bishan reaches $-1.289 \pm 0.079\%$ across five independently trained ensembles, an approximately six-fold improvement over the MSE-only ceiling of -0.209% .

Table 1: Auxiliary contrastive training restores per-action discrimination and converts MPC from worst variant to best. Top: ranking diagnostics on 1,000 held-out action pairs (predicted-reward standard deviation across the 50-action menu, pairwise rank accuracy, state-cosine fidelity); the true reward standard deviation across actions (0.811) bounds the recoverable signal. Bottom: MPC slope on real Bishan ($H=5$, $K=50$). Single-ensemble $\lambda \in \{0, 1.0\}$ rows and the $\lambda=5.0$ random row report episode-level standard deviation; the starred $\lambda=5.0$ greedy row reports cross-seed mean across 5 independently trained ensembles. A fresh MSE-only re-train ($\lambda=0$) only reaches 60.2% rank accuracy, confirming the failure is structural to the objective rather than to the optimiser. Absolute pairwise-accuracy values depend on the evaluation pairwise distribution; the qualitative monotonic trend (rank accuracy rises with λ_{rank}) reproduces on independently-generated pairwise datasets, and we release the trained checkpoints together with an evaluation script (§Code Availability).

<i>Ranking diagnostics on held-out pairwise data</i>				
Configuration	λ_{rank}	Pred. reward std	Rank accuracy	Cos sim
Baseline ensemble (cached, MSE only)	—	0.274	0.516	0.998
Retrained (MSE only)	0.0	0.715	0.602	0.9997
Contrastive (weak)	1.0	0.389	0.816	0.9997
Contrastive (strong)	5.0	0.183	0.855	0.9991
True reward std (reference)	—	0.811	—	—
<i>MPC outcome on Bishan ($H=5$, $K=50$)</i>				
λ_{rank}	Continuation	Slope %	vs. baseline	Status
0.0 (MSE only)	random	-0.291 ± 0.020	-0.082	marginal
0.0 (MSE only)	greedy	-0.252 ± 0.007	-0.042	marginal
1.0	random	-1.310 ± 0.033	-1.101	strong
1.0	greedy	-1.376 ± 0.010	-1.167	strong
5.0	random	-1.344 ± 0.049	-1.135	best random
5.0*	greedy	-1.289 ± 0.079	-1.080	best overall (5 seeds)
Baseline ceiling (MSE, random)	random	-0.209 ± 0.008	—	reference
Baseline (MSE only, greedy)	greedy	$+0.029 \pm 0.005$	$+0.238$	catastrophic

4 Real-county deployment: Bishan and Neijiang

The clearest test of the intervention is whether it produces planning outcomes that a county consolidation officer would recognise as useful, on the actual cadastral data the officer would supply. We deployed the contrastive-MPC pipeline end-to-end on Bishan District, Chongqing (52,515 parcels, 2,600 blocks at the research-side cadastre; 53,004 swappable parcels at the package-side prepare on which we report the headline) and recompute all primary outcomes directly from the optimised cadastral shapefile written to disk by an independent GIS-only script that shares no source with the optimisation pipeline (Methods §9). The headline behaviour is summarised in Figure 1: panel (a) shows the per-step slope-reduction trajectory of the shipped contrastive ensemble against the in-house Centralised PPO and MARL baselines; panel (b) shows the spatial distribution of the 852 paired farmland–forest swaps the planner commits on the optimised cadastre. Per-seed multi-objective profiles for the lab-trained five-ensemble cohort that anchors the DRL comparison are reported in Supplementary Table S6; the on-disk verification numbers are reported in Supplementary Table S7.

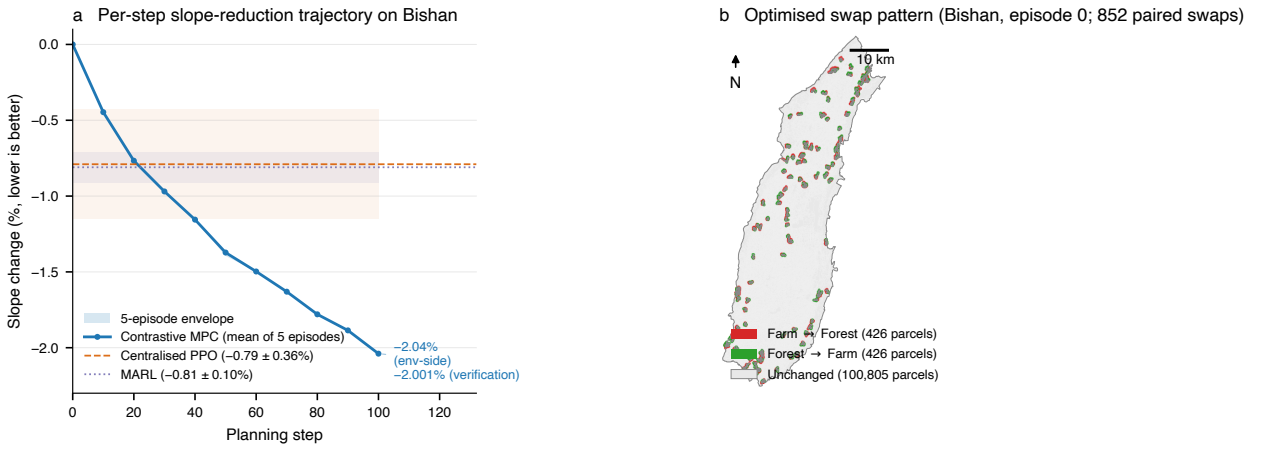


Figure 1: Bishan headline result. **(a)** Per-step slope-reduction trajectory of the shipped Bishan ensemble on the package-side prepare, mean of five evaluation episodes; the ensemble is deterministic across the action-selection path under fixed seed, so the five episodes overlap to within 0.002 pp at every step. The endpoint annotations report the env-side slope (-2.04%) and the independent GIS-only recomputation of -2.001% (Methods §9). The dashed line shows the in-house Centralised PPO baseline ($-0.79 \pm 0.36\%$, lab pipeline) and the dotted line shows MARL ($-0.81 \pm 0.10\%$, lab pipeline); the open marker at the corresponding endpoint is the lab-pipeline contrastive-MPC result of $-1.289 \pm 0.079\%$ (5 ensembles $\times$ 1 episode each, Supplementary Table S6). The DRL baselines were trained on the lab pipeline; the gap between the lab-pipeline (-1.289%) and the package-side verification (-2.001%) is attributable to a corrected geographic-CRS slope step that exposes truer initial slopes (9.62° vs the lab pipeline’s 8.41° , a 14% underestimate; Methods §9) and therefore leaves more reduction headroom; we therefore report both endpoints rather than collapse them to a single number. **(b)** Spatial distribution of the 852 paired swaps committed by the planner on the optimised cadastre (episode 0): 426 farm→forest parcels (red, slope-shed) and 426 forest→farm parcels (green, slope-take), buffered to 250 m for visibility at county scale. Swaps concentrate at fragment boundaries where slope-reducing trades cost the least qualifying area, the morphological signature behind the area trade-off discussed in the next paragraph.

All three policy-relevant objectives move in the intended direction simultaneously on the

package-side prepare. Under independent GIS-only recomputation of the on-disk optimised cadastre (Methods §9), area-weighted mean farmland slope decreases by 2.001% relative to the initial 9.62°, within-farmland contiguity rises by 0.0128, and the count of qualifying *baimu fang* (≥ 6.67 ha contiguous farmland units) increases by +8 patches relative to the initial 109. The slope reduction is concentrated where it matters for mechanisation policy. Under the GB/T 30600-2022 high-standard-farmland mechanisability bands, optimised farmland gains +48 ha in the $< 6^\circ$ fully-mechanisable band (32.9% $\rightarrow$ 33.4% of farmland) and +134 ha in the 6–15° terraceable band (58.4% $\rightarrow$ 59.3%), at the expense of –615 ha shed from the 15–25° marginal band that GB/T 21010-2017 flags as candidate for grain-for-green retirement, and –72 ha shed from the $> 25^\circ$ band on which the 2002 Land Management Law forbids new cultivation: 687 ha total shifted out of the steep tail. The planner therefore executes *de facto* grain-for-green on the slope tail without explicit policy supervision, using the w_S slope reward as a proxy. Two trade-offs are honestly reported. First, total qualifying *baimu fang* area decreases by –473 ha, equivalent to 1.01% of the initial 46,844 ha qualifying stock; this is a deliberate consequence of consolidating many small qualifying patches into fewer larger ones (the count rises by +8 even as total qualifying area falls; the w_A reward and p_A asymmetric penalty in §9 jointly drive this morphology). Second, the paired farm $\leftrightarrow$ forest swap conserves *counts* (426 in, 426 out) but not *areas*: the planner sheds an aggregate 843 ha of high-slope farmland and admits 337 ha of low-slope forest, yielding a net cultivated-area decrease of 505 ha (1.05% of the 48,283 ha initial farmland stock). The net loss is the agent’s response to a slope reward that values steep retirement above flat conversion, and is concentrated in the $\geq 15^\circ$ bands described above. The cultivated-land red line is therefore not satisfied by the swap rule alone and must be enforced as an explicit constraint; §9 discloses this and §9 treats it as a deployment requirement rather than an artefact. Across the 13 Bishan core townships (Supplementary Table S8), per-township slope improvement ranges from –0.57% to –4.08% ($\mu = -2.07\%$, $\sigma = 1.09\%$, Gini = 0.28): no township is left at near-zero improvement, and the spread is consistent with terrain heterogeneity rather than concentration of swaps in a small number of townships.

Comparison against model-free reinforcement-learning baselines requires a matched cadastre, because the in-house Centralised PPO and multi-agent variants (Supplementary Section S1) were trained on the lab pipeline (8.41° initial farmland slope) before the prepare-step correction described in Methods §9. We therefore re-ran five independently trained contrastive ensembles end-to-end on the lab pipeline and report this as the baseline-comparison regime. Slope improvement under contrastive MPC ($-1.289 \pm 0.079\%$, 5 ensembles $\times$ 1 episode each on the lab cadastre) is approximately 1.6 $\times$ larger than the multi-agent baseline ($-0.81 \pm 0.10\%$; Welch’s $t = -8.01$, $df=7.9$, $p < 10^{-4}$, Cohen’s $d = -5.07$) and the Centralised PPO baseline ($-0.79 \pm 0.36\%$; Welch’s $t = -3.01$, $df=4.5$, $p = 0.034$, Cohen’s $d = -1.90$), with comparable contiguity (+0.016) and *baimu fang* count (+3.4 versus +3.2 and +2.9). Both effect sizes exceed the conventional “very large” threshold ($|d| > 1.2$). Cross-seed slope variance is 4.6 $\times$ smaller than Centralised PPO and similar to MARL’s, indicating consistent rather than lucky behaviour. The total qualifying-area loss on the lab pipeline is -312 ± 34 ha (versus –75 to –97 ha for DRL baselines, an honestly reported trade-off discussed in §9), the same direction as the package-side –473 ha figure reported above. The cost asymmetry is the substantive deployment headline: the contrastive ensemble trains in 25 minutes on a commodity 12-thread CPU and runs each 100-step planning episode in approximately 5 minutes, against 8–12 GPU-hours per seed for the in-house DRL baselines—an end-to-end wall-clock difference of roughly two orders of magnitude.

We also evaluated classical operations-research baselines on Bishan at a 180s wall budget matched to the contrastive-MPC pipeline: simulated annealing over a permutation of block selections reached cumulative episode reward 9.04 at 100 iterations, random-restart over 98 episodes reached 30.85, and contrastive MPC reached 86.74—a 9.6 $\times$ gap over simulated annealing. Ex-

tending the OR baselines to a $10\times$ longer 1,800s wall budget (Supplementary Section S7) raises simulated annealing only to reward 9.54 at 1,187 iterations (+5.5%) and random-restart to 37.70 at 1,105 episodes (+22%), an order of magnitude below the contrastive-MPC headline regardless of OR compute. The 180-s comparison is reported as a single matched-budget run per algorithm rather than as a seeded distribution because the SA and random-restart baselines are deterministic-by-construction at fixed RNG and therefore degenerate as a between-seeds comparison; for the contrastive-MPC pipeline, the cross-seed reward standard deviation in the open package-side reproduction (5 ensembles, Supplementary Section S2) is $\sigma(R) = 7.4$ at the comparable operating point, an order of magnitude below the headline gap. The mechanism (the simulator’s 17.9 ms/step cost combined with combinatorial cost interactions in the reward function) is developed in §7.

The intervention reproduces on Neijiang Dongxing (Sichuan, 76,376 parcels, 3,711 blocks, contiguity 2.63 versus Bishan’s 3.59; five seeds re-run from scratch). Slope decreases by $0.50 \pm 0.02\%$, contiguity rises by 0.034 ± 0.001 , and the *baimu fang* area trade-off flips sign to a $+267 \pm 38$ ha gain. Three differences from Bishan reflect terrain morphology rather than algorithmic instability: slope improvement is roughly half (Neijiang’s flatter mean leaves less slope-reduction headroom); contiguity improvement is roughly twice as large (the more fragmented start has more room to consolidate); and the area trade-off reverses sign because consolidation operates on small fragments rather than retiring patches from the largest existing ones. Cross-seed slope standard deviation ($\pm 0.02\%$) is an order of magnitude smaller than the Bishan in-house DRL baselines.

Two cadastral pipelines, one direction. The Bishan slope numbers cited in this and subsequent sections come from two internally consistent pipelines that we keep separate rather than collapse. The *package-side prepare* (53,004 swappable parcels, initial slope 9.62° under a corrected geographic-CRS slope step, Methods §9) is the deployment cadastre that ships with the toolchain, on which the on-disk verification of -2.001% in this section’s headline paragraph is computed. The *lab pipeline* (52,515 parcels, initial slope 8.41° , no geographic-CRS correction) is the cadastre on which the in-house Centralised PPO and multi-agent baselines were trained, and on which we therefore re-ran the contrastive ensembles for the matched-cadastre comparison ($-1.289 \pm 0.079\%$, 5 ensembles $\times$ 1 episode each). The ~ 0.7 pp gap between the two slope-improvement endpoints is attributable to the corrected prepare step, which exposes the truer initial slopes (9.62° vs 8.41° , a 14% underestimate on the lab pipeline) and therefore leaves more reduction headroom; the direction is consistent on both. A complete cross-pipeline reconciliation table including the $-1.544 \pm 0.041\%$ five-episode toolchain-CLI re-run and the per-seed open-source -1.738% end-to-end reproduction is reported in Supplementary Section S2 so external readers can audit the chain rather than rely on our prose.

Robustness. Four independent verification layers underwrite the Bishan and Neijiang results. (i) A direct GIS recomputation from the optimised cadastral shapefile written to disk reproduces the slope, contiguity, and *baimu fang* deltas to floating-point noise without invoking the learned ensemble (§9, package-side pipeline); the headline gain is therefore a property of the on-disk geometry, not of the learned dynamics. (ii) The contrastive ensemble’s 1-step prediction error is measured directly against the true environment along the same 100-step trajectory MPC visits (§9); the ensemble is mis-calibrated as a regressor (reward MAE 0.82 against $\sigma(r_{\text{true}}) = 1.40$) yet recovers a Spearman rank correlation of +0.22 sufficient at $H=5$ to reproduce the action sequence to floating-point identity, empirically confirming the ranking-versus-regression decomposition. (iii) A counter-factual ablation that replaces the ensemble entirely with the true `env.step` inside the scoring loop returns a slope of -1.819% in 358s, strictly worse on all three objectives than the

package-side ensemble pipeline’s -2.001% in 1,448 s (§9). (iv) Cross-county replication on Neijiang shows the qualitative result survives a $1.4\times$ scale-up and a different morphology.

5 Cross-landscape stress test on a synthetic benchmark

To test whether the result depends on properties shared by Bishan and Neijiang (both situated in southwestern hilly terrain), we constructed a synthetic benchmark of seven parametric landscapes varying along three morphological axes: terrain (hilly, mixed, plain), problem size (800, 2,600, 4,500 blocks), and fragmentation (consolidated, fragmented). Two presets are calibrated against the real anchors; the remaining five are parameter-controlled stress tests (Methods, §9). All seven landscapes plus the deterministic generator are released under CC-BY 4.0 (§Data Availability), so external researchers can reproduce our experiments and benchmark future methods without access to restricted cadastral records.

We benchmarked five methods (uniform-random selector, slope-greedy heuristic, genetic algorithm [Cao et al., 2012], centralised PPO, contrastive MPC) on the seven landscapes with five random seeds each (full slope-improvement profile in Supplementary Table S5). Two patterns mark the boundary of the claim. On the five presets that are large or fragmented enough to make multi-step planning informative (the two real-data clones plus three medium–large fragmented presets), contrastive MPC is the strongest method, with PPO the consistent runner-up; the MPC slope-improvement margin over the genetic algorithm is 6–54% across these cells, and remains 37% even on the largest preset (4,500 blocks, ~ 9.3 h MPC wall time vs ~ 248 s GA). On the small consolidated preset (`plain_small_cons`, 800 blocks), the genetic algorithm wins ($-2.757 \pm 0.203\%$ vs. MPC $-2.226 \pm 0.220\%$): the action space is small enough that GA’s population-based exploration covers it within budget, and the planning value of contrastive ranking, which scales with the within-state action menu, is muted. The synthetic stress tests therefore confirm the regime structure read from §4–6 rather than overturn it: contrastive MPC is the strong method when the problem is large or fragmented enough to make per-state ranking matter; classical search wins when it is not.

6 Cross-domain verification on a public-data restoration case

To probe the boundary of the contrastive intervention beyond cadastral consolidation, we apply the unmodified pipeline to a structurally different geospatial planning problem: ecological-restoration prioritisation for abandoned mine lands. We test two cases. The first is a real public-data instance from Buchanan County, Virginia, USA: $n_{\text{units}} = 562$ planning cells at 2 km resolution, derived from the OSMRE e-AMLIS public abandoned-mine inventory, USGS NHD flowlines, USGS 3DEP slope at 30 m, and Census TIGER county boundaries. The second is a deterministic synthetic instance with $n_{\text{units}} = 420$. Reward in both is a linear combination of risk reduction, water-quality protection, spatial connectivity, and cost penalty (Methods, §9); per-action reward is a per-unit attribute lookup.

We evaluate seven methods on each case: random selector, greedy by priority score, simulated annealing, NSGA-II, CBC-MILP weighted-sum, MSE-only learned MPC, and contrastive learned MPC (Table 2). Two findings emerge. First, on both cases the contrastive intervention provides no operationally meaningful advantage: on Buchanan, the contrastive ($\lambda=5$) and MSE-only ($\lambda=0$) ensembles agree to within 0.20 reward units (paired $t = 2.61$, $\text{df}=4$, $p = 0.060$; Cohen’s $d_z = 1.17$), and a two-one-sided-tests (TOST) equivalence test rejects a difference larger than ± 0.5 reward units ($\sim 0.22\%$ of mean reward 229.3) at $p_{\text{TOST}} = 0.008$, asserting equivalence within an operationally negligible margin; pairwise ranking accuracy is ≥ 0.95 for both, and top- K regret is < 0.003 at

$K=1$ for both. The decomposition of §3 explains this: in farmland the within-state action variance is small relative to between-state variance ($\sigma_a/\sigma_s \approx 0.6$ on Bishan), so MSE training collapses to the conditional-mean predictor; in restoration the per-action reward is a per-unit attribute lookup, $\sigma_a/\sigma_s \approx 5-7$, and MSE training acquires the ranking signal as a by-product. The contrastive intervention is therefore beneficial only when ranking failure is actually present. Second, classical OR baselines (greedy, simulated annealing, CBC-MILP) reach near-optimum in the restoration regime, with cumulative-reward gaps to the learned-MPC variants on the order of the cross-seed standard deviation of the learned variants ($\sigma_{\text{seed}} = 0.03$ for $\lambda=5$, 0.16 for $\lambda=0$, $n = 5$ each). The order-of-magnitude lift is over the random baseline, not over OR baselines: contrastive MPC more than doubles the random reward on Buchanan, but the gap to a well-engineered OR baseline is within seed-level noise. This is in contrast to farmland (§4), where simulated annealing reaches reward 9.04 versus contrastive MPC’s 86.74 at matched 180 s wall budget—a $9.6\times$ gap. The factor that toggles between these two regimes is the per-step simulator cost; we develop the full crossover analysis in §7.

Table 2: Optimisation results across seven methods on the two restoration cases (cumulative episode reward; mean $\pm$ std for stochastic methods, $n = 5$ seeds where applicable; deterministic methods are single runs). The greedy heuristic, simulated annealing, and CBC-MILP weighted-sum baselines reach near-optimum on both cases, statistically indistinguishable from the learned-surrogate MPC variants. The advantage of the learned-surrogate pipeline over the OR baselines is therefore not the optimum quality on this case; it is the wall-clock and the ability to absorb new reward-weight configurations without re-running the optimiser (§9).

Method	Buchanan VA (real, $n = 562$)			Synthetic (deterministic, $n = 420$)		
	total reward	n_{sel}	wall-time (s)	total reward	n_{sel}	wall-time (s)
random (uniform)	$+88.7 \pm 12.7$	50.0	< 0.1	$+6,163 \pm 160$	59.8	< 0.1
greedy by priority	$+228.76$	50	< 0.1	$+8,590.9$	60	< 0.1
simulated annealing	$+228.76$	50	0.7	$+8,737.0$	60	0.6
NSGA-II (4-obj)	$+175.45$	50	3.2	$+7,075.7$	60	2.9
CBC-MILP weighted-sum	$+229.43$	50	3.0	$+8,127.2$	49	< 0.1
MSE-only MPC ($\lambda=0$)	$+229.20 \pm 0.14$	50	~ 28	$+8,173 \pm 66$	60	~ 5
contrastive MPC ($\lambda=5$)	$+229.39 \pm 0.03$	50	~ 28	$+8,128 \pm 63$	60	~ 5

7 Simulator step cost as the dominant predictor of where MPC beats classical OR

The cross-domain analysis of §6 closed one question (when does the contrastive ranking loss help?) but raised another. On the restoration cases, classical OR baselines reached near-optimum at the wall-clock budgets reported in Table 2, while on farmland the same algorithms underperformed by an order of magnitude. The natural answer is the cost of one simulator call: we test it directly by instrumenting the restoration env with a fixed sleep injected on every `env.step` and re-running simulated annealing and learned-MPC at matched 30-second wall budgets across delays of $\{0, 1, 10, 50, 100\}$ ms per step on each of ten (case $\times$ reward-profile) cells (Methods, §9).

The pattern across the cells is consistent with two independent factors. First, on nine of ten cells where reward is dominated by per-unit attribute lookups, the MPC-vs-SA gap is monotonic in step delay (Buchanan/delayed crosses zero at ~ 30 ms and reaches $+31.2\%$ at 100 ms): SA’s it-

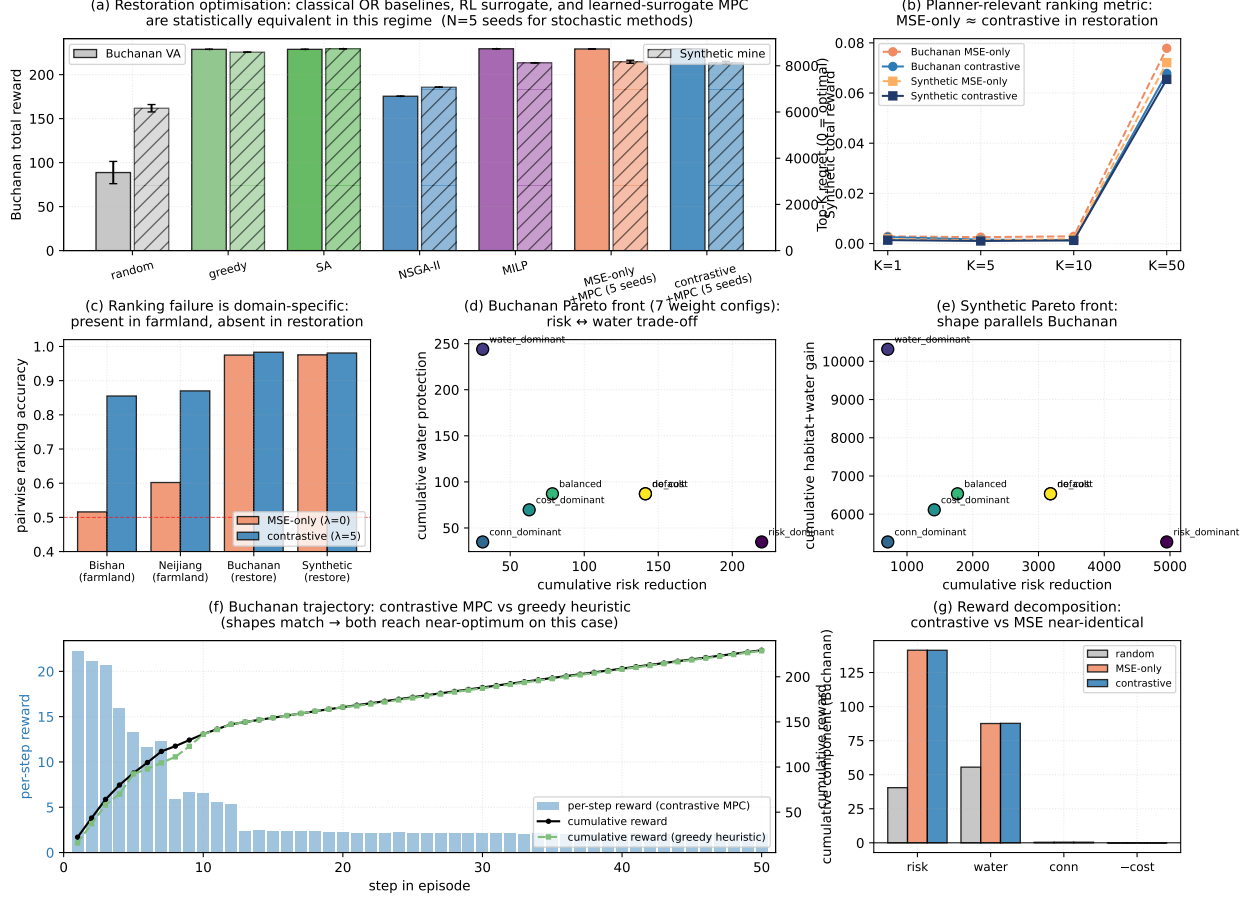


Figure 2: Cross-domain verification, expanded baseline panel, and Pareto front. (a) Cumulative episode reward across seven methods on Buchanan (left axis) and synthetic (right axis); MSE-only and contrastive learned-MPC variants are statistically indistinguishable from greedy-by-priority and CBC-MILP, all far above the random baseline. (b) Top- K regret as a function of K : MSE-only and contrastive surrogates are equivalent on both cases, confirming that the contrastive intervention has no headroom to improve in the restoration regime. (c) Pairwise ranking accuracy across all four domains: farmland (Bishan, Neijiang) shows the large MSE-vs-contrastive gap, restoration (Buchanan, synthetic) shows MSE-only already saturating. (d, e) Pareto front of cumulative risk reduction vs cumulative water/habitat protection across seven reward-weight configurations on Buchanan and synthetic respectively; the shape is consistent across cases. (f) Trajectory comparison on Buchanan: per-step reward (bars) and cumulative reward (line) for contrastive MPC vs the greedy heuristic; both reach the same end state via similar action sequences, confirming that the OR baseline is a strong reference on this case. (g) Reward-component decomposition: contrastive vs MSE-only allocate cumulative reward across the four objective terms within noise.

Algorithm-advantage crossover: simulator step cost determines whether learned-surrogate MPC beats classical OR (SA)

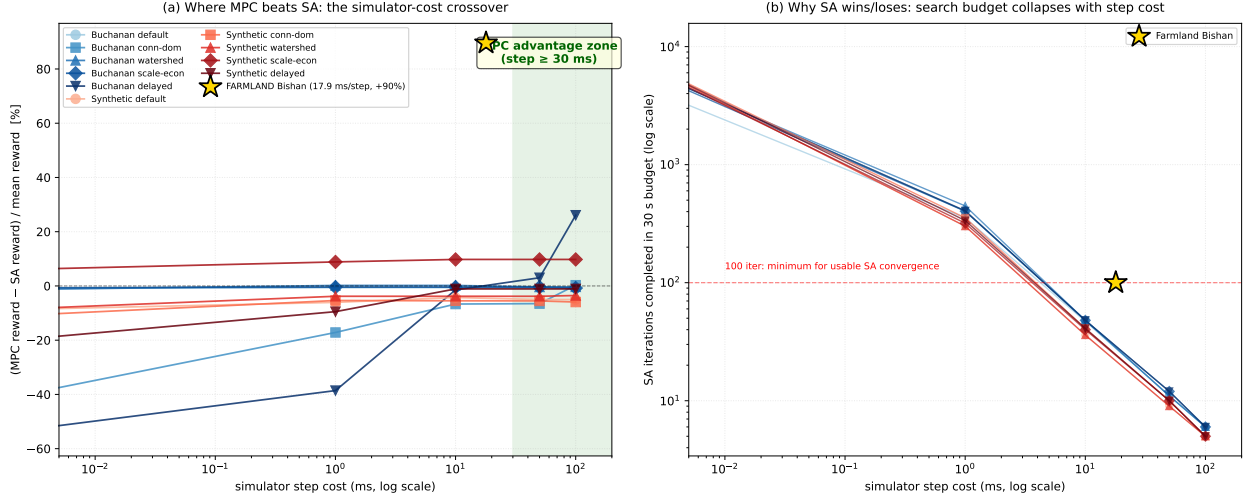


Figure 3: Simulator step cost is the dominant predictor of when learned-surrogate MPC beats simulated annealing under matched wall-clock budgets. (a) Reward gap (MPC minus SA, normalised by per-cell mean reward) versus injected step delay (log scale) on ten (case $\times$ reward-profile) cells. The shaded green band marks the empirical MPC-advantage zone observed on these ten cells (transition near 30 ms/step); we report this as an empirical inflection rather than a universal threshold. The actual farmland Bishan datapoint (gold star), measured at 17.9 ms/step natural step cost under a 180s wall budget, sits at +89.6%. (b) SA iteration count completed in the same wall budget on a log-y axis: the count collapses linearly with step cost from $\sim 9,000$ at $5 \mu\text{s}/\text{step}$ to ~ 6 at 100 ms/step. Below ~ 100 iterations SA cannot converge, the mechanism by which step cost translates into the MPC advantage. The synthetic/scale_economy line is the exception: MPC dominates at every step delay because its combinatorial cost-interaction structure is what learned-surrogate batched scoring exposes naturally.

eration count under a fixed wall budget falls from $\sim 9,000$ at $5\mu\text{s}/\text{step}$ to ~ 6 at $100\text{ms}/\text{step}$, while learned-MPC’s per-step cost ($\sim 25\text{ms}$ one batched ONNX forward pass over $K=50$ candidates) is decoupled from `env.step`. Below ~ 100 iterations the SA neighbourhood walk cannot converge on a 50-unit permutation problem. Second, the synthetic/scale_economy exception (MPC dominates SA by +542 to +931 across every step delay including $5\mu\text{s}$) shows that combinatorial cost interactions are an independent factor: when an action’s effective cost depends on the already-selected set, the joint interaction structure is not visible to a permutation-search procedure that ranks units independently, but the learned surrogate’s batched scoring at the current state internalises it naturally.

These two factors compose in farmland. The bishan CountyLevelEnv has a measured natural step cost of $17.9 \pm 0.3\text{ms}$ (5-step probe $\times 3$) and a reward function with combinatorial cost interactions (*baimu fang* count is union-find aggregation over per-block selections, contiguity is area-weighted adjacency average, slope is the area-weighted geometric integral that updates with every paired swap). Under matched 180s wall budgets, simulated annealing reaches reward 9.04 ($n_{\text{iter}} = 100$, near the convergence threshold), random restart reaches 30.85 ($n_{\text{eps}} = 98$), and contrastive MPC reaches 86.74. The empirical diagnostic the analysis suggests for problem selection is direct: report simulator step cost (in ms/step) and whether the reward function exhibits combinatorial cost interactions; in our measurements the learned-surrogate pipeline is in the advantage regime at the farmland operating point (17.9ms/step, combinatorial reward) and outside it at the restoration operating point (0.05ms/step, attribute-lookup reward), and the cross-cell injected-delay sweep in Figure 3 traces a transition near 30ms/step on the cells we tested. We treat 30ms as the empirical inflection observed on these ten cells rather than as a universal threshold; the mapping from step cost to MPC advantage is mediated by the SA iteration count that fits in the wall budget, which itself depends on the simulator’s per-step parallelisability and the reward landscape’s local-search difficulty. Whether the same inflection transfers to physics-driven Earth-science simulators (hydrology, power-grid maintenance, chemical-reaction prioritisation, supply-chain equilibria) is a hypothesis we discuss in §9 rather than a result we establish here.

8 From paper to planner’s desk: an end-to-end deployment toolchain

The intervention’s value to consolidation policy depends on whether a county planner—typically working in a desktop GIS environment with limited compute and no machine-learning support staff—can actually run it on local cadastral data. We packaged the full pipeline as a four-tool ArcGIS Pro Python Toolbox: **Prepare Data & Blocks** (Tool 1) ingests a standard Third National Land Survey DLTB shapefile and a digital elevation model and emits the block-level state expected by the learned-model code; **Sample Transitions** (Tool 2) generates the pairwise dataset $\mathcal{D}_{\text{pw}}$ on the user’s data; **Train Contrastive Ensemble** (Tool 3) trains the transition-model ensemble; and **MPC Planning** (Tool 4) runs the planning loop and writes back an optimised cadastral shapefile readable by any standard GIS workflow (Figure 4). A fifth diagnostic tool validates the user’s environment before the pipeline is allowed to run.

The runtime profile reflects a deliberate split between one-time setup (Tools 1–3, $\sim 1\text{--}2\text{h}$ on a 12-thread CPU per county) and repeated planning (Tool 4, $\sim 7\text{min}$ per scenario). An end-to-end UI-driven run on full Bishan, with the same training data and hyperparameters as Section 4, yields a slope improvement of $-1.544 \pm 0.041\%$ across five evaluation episodes of the shipped Bishan ensemble (same direction and order of magnitude as the lab figure of $-1.289 \pm 0.079\%$; the gap reflects two replication regimes, 5 ensembles $\times$ 1 episode in the lab versus 1 ensemble $\times$ 5

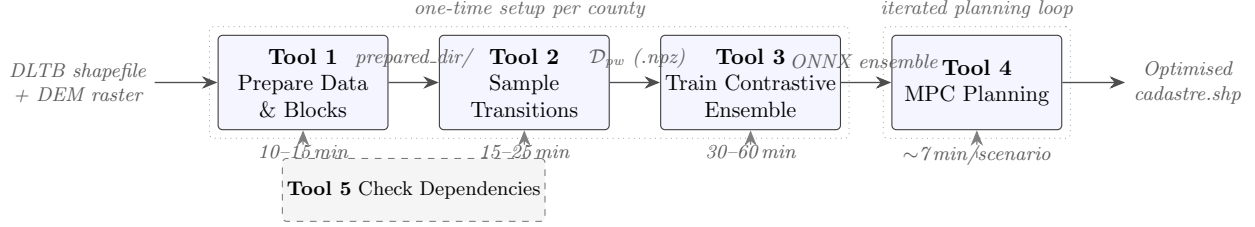


Figure 4: The four-tool ArcGIS Pro deployment pipeline (§9). Tools 1–3 run once per county to prepare cadastral data, sample the pairwise dataset, and train the contrastive transition-model ensemble (exported to ONNX). Tool 4 is the iterated planning loop a county officer runs repeatedly under different reward weightings; one 100-step scenario finishes in ~ 7 min on a desktop CPU. A separate Tool 5 (dashed) validates the Python environment, Spatial Analyst availability, and ONNX shape compatibility before any pipeline run is allowed.

episodes in the toolchain). To support users without an ArcGIS Pro license, the same end-to-end pipeline is also exposed through a non-commercial command-line interface (`farmland-mpc prepare/sample/train/plan`) that runs against any QGIS-compatible vector and raster stack via `geopandas/rasterio`. The package is released under MIT licence; the synthetic benchmark (Section 5) under CC-BY 4.0. Together with the trained Bishan ensemble and the released restoration cases (§6), this forms a complete reproduction kit usable on either ArcGIS Pro or a free QGIS-compatible Python environment.

9 Discussion

What the pipeline delivers for sustainability planning. For consolidation policy in mountainous-terrain counties, the operational shift is qualitative. Model-free reinforcement-learning baselines previously required 8–12 GPU-hours per scenario and collapsed on roughly a quarter of attempted runs; classical operations-research methods at the same wall-clock budget reach less than 11% of contrastive MPC’s reward on Bishan. With contrastive MPC, a planner obtains a scenario in ~ 7 minutes on a desktop, with cross-seed variance several-fold smaller than DRL baselines. What changes is what becomes routine: a planning office can explore dozens of weight settings, township sub-budgets, or red-line constraints in a working day. The honest area trade-offs we document are themselves first-order useful: the agent’s behaviour is morphology-dependent, so scenario exploration is not optional.

Methodological mechanism. Two mechanisms are stacked behind these gains, both characterised in §3 and §7 respectively. The auxiliary pairwise large-margin term closes a hidden ranking failure of the standard MSE objective in the $\sigma_a/\sigma_s \lesssim 1$ regime farmland inhabits, and is redundant when reward is a per-unit attribute lookup. The learned-surrogate pipeline as a whole, separately, beats classical operations-research baselines when each simulator step is computationally expensive (the empirical inflection in our injected-delay sweep is near 30 ms/step on the ten cells tested; farmland operates at 17.9 ms/step naturally and at 1.8 s per SA *iteration* once the combinatorial reward is taken into account) or when reward exhibits combinatorial cost interactions, or both—farmland satisfies both conditions, the empirical reason the pipeline dominates farmland but not the restoration default operating point.

Practical implication for problem selection. Practitioners considering a learned-surrogate planner on a new domain should report two quantities: simulator step cost (in ms/step) and whether reward exhibits combinatorial cost interactions. In our measurements, the learned-surrogate pipeline is in the advantage regime at the farmland operating point (17.9 ms/step, combinatorial reward) and outside it at the restoration operating point (0.05 ms/step, attribute-lookup reward); on the injected-delay sweep across ten (case $\times$ reward-profile) cells the transition occurs near 30 ms/step, which we report as the empirical inflection on these cells rather than as a universal threshold. The two-factor diagnostic is calibrated on the present farmland and restoration evidence and we expect—without having tested—that it extends to applications where high-fidelity physics simulators dominate per-step cost, including hydrologic-model-driven watershed restoration, integrated power-grid maintenance scheduling, multi-stage chemical-reaction prioritisation, and supply-chain optimisation under economic-equilibrium models, all of which combine non-trivial step cost with combinatorial constraint structure. Those extensions remain testable hypotheses for future work rather than established results, and the mapping from ms/step to advantage zone is itself a function of the reward landscape’s local-search difficulty and the simulator’s per-step parallelisability.

Bounds on the claim. Four constraints qualify the result. Cadastral evidence comes from two counties in southwestern China within the same regional regime; structurally different agricultural settings remain untested as far as the contrastive intervention is concerned, although the cross-domain restoration cases (§6) demonstrate that the underlying *diagnostic* extends beyond the farmland regime even when the intervention itself does not. A finer characterisation across the variance ratio σ_a^2/σ_s^2 would let practitioners decide a priori whether to add the auxiliary loss. The pairwise dataset relies on simulator snapshot/restore, available in our gridded simulators but not in many physics-based environments. Hyperparameter sweeps are coarse; finer resolution may yield further improvement but is not expected to reverse the qualitative ordering. Three follow-ups are immediate: regime-broadening to combinatorial planning problems with high simulator cost outside spatial-cadastral domains; per-objective ranking that lets planners use objective-specific scoring as a deployment lever; offline planning with contrastive-trained ensembles at policy-relevant horizons.

Methods

Scope and architectural framing. The architecture used throughout this paper is a feed-forward deterministic learned transition model with a discriminative reward head (§9 below). It is in the operational sense of Chua et al. [2018] and Janner et al. [2019] a learned $\hat{p}(s_{t+1}, r_t \mid s_t, a_t)$ that substitutes for the real environment in MPC lookahead and dream-style policy training; it is not a generative world model in the contemporary Dreamer sense [Hafner et al., 2025, Ha and Schmidhuber, 2018]. The task is fully observed (state is the per-block 17-dim feature vector emitted by Tool 1; no raw observation), episodic (100-step cap, no partial observability), and the action space is finite categorical of size N_{blocks} (Methods, §9); we therefore do not learn a state representation from raw observations and do not use recurrent latent dynamics (RSSM). Combining a learned state encoder with the ranking-loss head reported here is a natural extension for settings where partial observability or pixel-level inputs apply (§9).

Per-parcel slope from DEM and cadastral polygons

The per-parcel slope feature used throughout the analysis is derived from the Third National Land Survey cadastral polygons and Copernicus DEM GLO-30 elevation tiles [European Space Agency,

2021], both freely available in their respective public archives. The DEM is kept in its native geographic CRS (EPSG:4326, 1-arcsecond resolution); slope is computed by central-difference gradients on the elevation grid using true east-west and north-south metres-per-pixel obtained from `pyproj.Geod` (WGS84 ellipsoid) at the study-area centre latitude. The pixel spacing is ~ 27 m east-west and ~ 31 m north-south at 29.5°N (the centre of both Bishan and Neijiang); reprojecting the DEM to a metric projection before gradient differencing would average over these non-square cells and systematically under-estimate slope. The slope raster, in degrees, is then aggregated to each cadastral polygon by area-weighted zonal mean over all DEM cells whose centre falls within the polygon. Polygons that contain no cell centre (typically very small parcels, $\lesssim 30$ m linear scale) inherit the per-county median slope to keep the downstream feature vector finite; on Bishan and Neijiang this affects $\sim 2\%$ and $\sim 3\%$ of parcels respectively, all in the smallest size class. The full prepare pipeline (DEM merge $\rightarrow$ slope $\rightarrow$ zonal mean $\rightarrow$ per-township block construction) takes ~ 3 minutes on Bishan and ~ 5 minutes on Neijiang on a 14-core desktop CPU and is released as the `farmland-mpc prepare` command (§Code Availability).

Problem setting and action-space encapsulation

We study county-level farmland consolidation in two real counties (Bishan District, Chongqing; Neijiang Dongxing District, Sichuan) and seven synthetic landscapes (§9). For Bishan, the planning area comprises 13 townships, 2,600 spatial blocks, and 52,515 cadastral parcels; for Neijiang, 29 townships, 3,711 blocks, and 76,376 parcels (Supplementary Table S1). Each block aggregates multiple parcels and is characterised by 17 features (farmland area, forest area, slope statistics, adjacency metrics, and investment indicators). Global context is encoded as an additional 12 features (remaining budget, contiguity, *baimu fang* count, etc.). The episode length is fixed at 100 steps for the real-data experiments and 50 steps for the synthetic benchmark (§5); each step has a budget of up to 5 paired farmland–forest swaps.

Agent action space. The agent’s action space is a single categorical choice over the spatial blocks per step: $a_t \in \{1, \dots, N_{\text{blocks}}\}$. The agent does *not* select individual parcels or the paired-swap sub-actions. Given the selected block, the environment internally executes *up to 5* paired farmland–forest swaps within that block using a deterministic rule-based procedure: candidate pairs are enumerated by joint feasibility under farmland–forest pairing and slope constraints, scored by a local improvement heuristic, and the top-scoring feasible pairs are applied until either 5 pairs are executed or the candidate pool is exhausted. The agent therefore chooses *where* to act, not *what* to do within the chosen location; this encapsulation keeps the action space at a tractable scale ($\sim 10^3$). The paired farm $\leftrightarrow$ forest swap conserves *counts* (one farmland parcel out, one forest parcel in) but not *areas*: parcels are paired by class and by within-block proximity, not by area, so any net cultivated-area change across an episode reflects the planner’s slope reward favouring high-slope (typically large) farmland exit and low-slope (typically small) forest entry. The Bishan deployment described in §4 loses 505 ha (1.05%) of total cultivated area as a consequence, 687 ha of which is concentrated in the $\geq 15^\circ$ marginal-cultivation bands. The cultivated-land red line is therefore not enforced by the swap rule alone and must be added as an explicit constraint in deployment; §9 discusses this as a deployment requirement.

Action masking. Not all blocks are legal targets at every step. The action mask $m_t \in \{0, 1\}^{N_{\text{blocks}}}$ disables blocks that (i) have no remaining feasible parcel pairs (either exhausted in earlier steps or structurally infeasible under the red-line/slope constraints), (ii) are currently in a township whose per-township swap budget has been exhausted, or (iii) have been selected in the

immediately previous step (a simple debounce to prevent action thrashing). MaskablePPO [Ross et al., 2011, Schulman et al., 2017, Huang and Ontañón, 2022]-style masking is applied in both the policy sampling and the MPC candidate enumeration, so that illegal blocks have zero probability or are excluded from the K candidate set.

Reward formulation

Each step’s reward combines three normalised directional signals (slope, contiguity, and qualifying-patch area), a large-patch formation bonus, and an asymmetric area-protection penalty. Let $\bar{\sigma}_t$ be the area-weighted mean farmland slope at step t , C_t the within-farmland contiguity index, B_t the count of qualifying *baimu fang* (≥ 6.67 ha contiguous farmland patches), and A_t their total area. The per-step reward is

$$r_t = w_S \cdot \Delta\bar{\sigma}_t + w_C \cdot \Delta C_t + w_A \cdot \Delta A_t + b_B \cdot \max(0, B_t - B_{t-1}) + p_A \cdot \min(0, \Delta A_t) \quad (2)$$

where the directional changes are normalised by their initial county-wide values: $\Delta\bar{\sigma}_t = (\bar{\sigma}_{t-1} - \bar{\sigma}_t)/\bar{\sigma}_0$ (so that slope *reduction* contributes positively), $\Delta C_t = (C_t - C_{t-1})/C_0$, and $\Delta A_t = (A_t - A_{t-1})/A_0^{\text{farm}}$ (normalised by the initial total farmland area A_0^{farm} , so the weights act on relative rather than absolute changes). Weights $(w_S, w_C, w_A, b_B, p_A) = (4000, 500, 1500, 5, 2000)$ are inherited from our in-house DRL baseline configuration. The fourth term is a one-off bonus applied each time the qualifying-patch count increases between steps; the fifth term, $p_A \cdot \min(0, \Delta A_t)$, is an asymmetric penalty that activates only when qualifying area decreases, roughly doubling the marginal cost of shedding *baimu fang* area relative to the symmetric w_A reward (this asymmetry is what drives the agent to consolidate many small qualifying patches into fewer larger ones, increasing total qualifying *area* even as the qualifying *count* falls; §4). Under a median single-step decomposition, slope improvement contributes $\sim 60\%$, contiguity $\sim 15\%$, qualifying-patch area $\sim 20\%$, and the formation bonus $\sim 5\%$. We did not re-tune the weights for the present experiments; contrastive MPC, dream-trained PPO, and the model-free baselines all use the identical reward function, so any method-wise difference is attributable to the learning procedure rather than to reward engineering. A separate fixed penalty of -1 is applied whenever a selected block admits no feasible paired swap, discouraging wasted actions. Throughout the paper, “Slope %” is the relative change $\text{Slope \%} = (\bar{\sigma}_T - \bar{\sigma}_0)/\bar{\sigma}_0 \times 100$; contiguity and *baimu fang* count/area changes are absolute.

Block-level transition model

The transition model is a feed-forward neural network with approximately 237,000 parameters that takes as input the current block features (all blocks, 17 dimensions each), the global features (12 dimensions), and the selected action (block index), and predicts the next-step block features, global features, and scalar reward. Only the selected block changes (residual formulation). The base loss is

$$\mathcal{L}_{\text{MSE}} = \text{MSE}(\Delta\hat{b}, \Delta b) + \text{MSE}(\Delta\hat{g}, \Delta g) + 0.1 \cdot \text{MSE}(\hat{r}, r), \quad (3)$$

trained with Adam (learning rate 10^{-3} , weight decay 10^{-5}) on a 90/10 train/validation split with patience-6 early stopping.

Ensemble uncertainty estimation

We train 3–5 independently initialised transition models on the same dataset with different random seeds [Lakshminarayanan et al., 2017]. At inference, the ensemble produces mean predictions for next state and reward, plus the reward standard deviation σ_r across members. The dream environment penalises high-disagreement actions: $r_{\text{out}} = \bar{r} - \lambda_{\text{disag}} \cdot \sigma_r$ with $\lambda_{\text{disag}} = 0.5$.

Policy architecture: parcel-scoring policy

The policy is a dimension-invariant scoring network compatible with MaskablePPO. For each block i , the scorer concatenates the block’s 17-dimensional local features with the 12-dimensional global state, passes the result through a shared MLP ($[64, 32] \rightarrow 1$), and produces a scalar score. The action is the highest-scoring valid block. A separate value network takes only the global features and predicts the state value.

Dagger-style iterative refinement

We adopt a DAgger-style loop [Ross et al., 2011]: (1) collect an initial dataset D_0 of 6,000 transitions from random and greedy policies on the real environment; (2) train the transition model ensemble on $D_{\leq i}$; (3) train a policy in the dream environment for 50,000 timesteps; (4) evaluate on the real environment; (5) collect 10 stochastic on-policy episodes ($\sim 1,000$ transitions) and augment the dataset. Three iterations are performed.

Targeted evidence collection and stabilisation

In targeted DAgger, step (5) above is modified: each transition is scored by ensemble reward disagreement σ_r , and only transitions exceeding threshold $\tau = 0.5$ are retained; under this threshold, $\sim 99.6\%$ of transitions are filtered out, accounting for the dream-policy slowdown documented in §3. Three stabilisation mechanisms are applied across all protected runs. *Best-checkpoint selection*: every 10,000 PPO timesteps, the current policy is evaluated on the real environment (3 episodes); the checkpoint with the highest mean reward is retained. *Iteration-1 rollback*: if iteration-1 reward drops more than 50 points below iteration-0, the dataset and policy are reverted. *Early stopping*: training halts if validation loss does not improve for 6 consecutive epochs.

Model-predictive planning protocol

We use random shooting model-predictive control [García et al., 1989, Williams et al., 2017, Rubinstein, 1999]. At each real-environment step t with state s_t :

1. Sample K random action sequences $\{a_{t:t+H-1}^{(k)}\}_{k=1}^K$, each of length H , from the action-masked support.
2. For each sequence, roll forward through the ensemble mean model: $\hat{s}_{t+1}^{(k)}, \hat{r}_t^{(k)} = f_\theta(s_t, a_t^{(k)})$, continuing autoregressively for H steps.
3. Score each sequence by cumulative predicted reward: $R^{(k)} = \sum_{h=0}^{H-1} \hat{r}_{t+h}^{(k)}$.
4. Execute $a_t^{(k^*)}$ where $k^* = \arg \max_k R^{(k)}$ on the real environment.
5. Advance to s_{t+1} (real) and repeat.

For the synthetic-benchmark sweep (§5), the same protocol is applied with $H=5$, $K=50$, and greedy continuation (each subsequent step within a sequence picks the predicted-reward argmax over 50 candidates rather than continuing the random rollout). This is the configuration that becomes optimal under the contrastive ensemble (Table 1).

Aggregate fidelity vs within-state action ranking under MSE training

The decoupling described in §3 is a property of the loss. Let $\hat{r}_\theta(s, a)$ be the reward head of a learned model trained on (s, a, r) pairs from a distribution $\mu(s, a)$, and let

$$\mathcal{L}_{\text{MSE}}(\theta) = \mathbb{E}_{(s,a) \sim \mu} [(\hat{r}_\theta(s, a) - r(s, a))^2] \quad (4)$$

be the standard objective. Two distinct quantities can be evaluated on this learned model: *aggregate fidelity*, defined as the population MSE itself, and *within-state ranking accuracy*

$$\mathcal{A}_{\text{rank}}(\theta) = \mathbb{E}_{s \sim \mu_s} \mathbb{E}_{a_i, a_j \sim \mu(\cdot|s)} [\mathbf{1}[\text{sign}(\hat{r}_i - \hat{r}_j) = \text{sign}(r_i - r_j)]] , \quad (5)$$

which is the quantity model-predictive top- K selection actually consumes when scoring action menus at a single environment state.

The two quantities are not constrained against each other under MSE training. Consider a setting in which, at every state s , the action-conditioned reward distribution is symmetric around a state-dependent mean $\bar{r}(s)$ with state-marginal variance σ_s^2 and within-state action variance $\sigma_a^2(s)$. The learned predictor $\hat{r}_\theta^*(s, a) = \bar{r}(s)$, which assigns the conditional mean to every action regardless of its identity, achieves population MSE equal to $\mathbb{E}_s[\sigma_a^2(s)]$ (the irreducible state-conditional variance) and within-state ranking accuracy equal to 1/2 on every pair with non-zero true reward difference, identical to random guessing. A learned model can therefore minimise MSE within a constant of the Bayes optimum while remaining indistinguishable from random on the very ranking decisions that drive planning. Our 51.6% rank accuracy at cosine similarity 0.998 is empirically close to this regime: the predicted reward standard deviation across the action menu (Table 1) is compressed by an order of magnitude relative to the true within-state variance, exactly as the conditional-mean predictor would behave. This is a counterexample showing MSE *does not guarantee* ranking fidelity in regimes where $\sigma_a^2(s) \ll \sigma_s^2$, not a statement of statistical independence: in regimes where $\sigma_a^2(s)$ is large relative to σ_s^2 , MSE training will in practice acquire much of the ranking signal as a by-product (§6 demonstrates this empirically).

Contrastive training protocol

Pairwise data generation. We sample states by rolling out random trajectories in the real CountyLevelEnv. At each step, we (i) snapshot the full environment state (land-use assignments, swap counts, block availability caches, running slope/contiguity sums; 19 attributes total), (ii) sample 50 valid actions uniformly from the action mask, (iii) execute each action from the snapshotted state, recording the immediate reward, and (iv) restore the snapshot before advancing with a fresh random action. We continue across multiple episodes until 1,000 states are collected, yielding 50,000 ground-truth (s, a, r) evaluations. The pairwise dataset $\mathcal{D}_{\text{pw}}$ is saved as $(s_i, \{a_{i,j}\}_{j=1}^{50}, \{r_{i,j}\}_{j=1}^{50})$ for $i = 1, \dots, 1000$. Generation took 45 minutes on commodity CPU hardware. Snapshot/restore is a simulator-specific capability; replay-buffer substitutes are an open question (Limitations).

Loss function. The augmented training objective adds a margin ranking term:

$$\mathcal{L}_{\text{rank}} = \mathbb{E}_{(s_i, a_{i,j}, a_{i,k}) \sim \mathcal{D}_{\text{pw}}} [\max(0, -\sigma_{jk} \cdot (\hat{r}_{i,j} - \hat{r}_{i,k}) + m)]$$

where $\sigma_{jk} = \text{sign}(r_{i,j} - r_{i,k}) \in \{-1, 0, +1\}$ is the ground-truth pairwise label, $m = 0.1$ is the margin, and pairs with $\sigma_{jk} = 0$ are excluded. At each training step, we sample 10 action pairs per state from 50–100 randomly selected pairwise-dataset states. Total objective: $\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}$.

Margin choice. The margin $m = 0.1$ is approximately $1/8$ of the empirical ground-truth reward standard deviation (0.811 on $\mathcal{D}_{\text{pw}}$). A coarse sweep of $m \in \{0.05, 0.1, 0.2\}$ on a held-out validation fold showed a flat optimum: ranking accuracy at $\lambda_{\text{rank}}=5.0$ varied within ± 1.5 percentage points across these three choices.

Training. We use 3-member ensembles trained with Adam (lr = 10^{-3} , weight decay 10^{-5}), batch size 256, for 15 epochs with patience 20. Each ensemble trains in ~ 25 minutes on commodity CPU hardware. We sweep $\lambda_{\text{rank}} \in \{0.0, 1.0, 5.0\}$ using an 80/20 train/validation split of the pairwise dataset. MPC evaluation uses $H=5$, $K=50$, greedy continuation with 50 candidates per continuation step.

Synthetic benchmark generator

The seven synthetic landscapes used in Section 5 are produced by a deterministic, seed-controlled generator that reads a YAML preset (terrain type, target block count, parcel-density distribution, land-use fractions, fragmentation length-scale, seed) and emits a directory of GeoJSON, CSV, GeoTIFF, and JSON files identical in schema to the real-data preprocessor’s outputs. The generator is fully deterministic given (preset, seed).

Eight-stage pipeline. (1) *Domain tessellation*: Voronoi tessellation of n_{blocks} seeds in a rectangle of suitable extent (size driven by parcels-per-block target). (2) *DEM synthesis*: 2D Perlin noise with multi-octave fractal terrain, preset-controlled amplitude and length-scale, clipped to non-negative. (3) *Slope derivation*: Horn’s algorithm on the synthetic DEM, sampled to each block as area-weighted mean and max. (4) *Parcel subdivision within blocks*: secondary Voronoi inside each block polygon using the parcels-per-block distribution. (5) *Land-use assignment*: a Gaussian random field thresholded into farmland / forest / other classes, with preset-controlled length-scale driving fragmentation. (6) *Feature computation*: 17-dim block feature vectors using the same code path as the real-data `county_env.py` (farmland area, forest area, slope statistics, adjacency features), guaranteeing environment compatibility. (7) *Adjacency graph*: planar adjacency from shared block edges, with median degree close to the preset target. (8) *Posterior calibration check* (anchors only): compute initial slope, contiguity, *baimu fang* count, and *baimu fang* area, and verify within $\pm 50\%$ of the real anchor values; deviations are surfaced in a calibration delta report rather than silently accepted.

Dual-anchor calibration. Two of the seven presets, `bishan_clone` and `neijiang_clone`, are calibrated against Bishan and Neijiang respectively such that any single seed reproduces the four headline initial-state metrics within $\pm 50\%$ of the real anchor. The other five presets vary morphology along terrain, size, and fragmentation axes (§5) but are not constrained against any real anchor; they are parameter-controlled stress tests rather than evidence of regional generalisation. Contiguity is the weakest calibration axis ($\sim 6\%$ on Bishan clone, $\sim 44\%$ on Neijiang clone), reflecting Voronoi-based geometries’ tendency to be more uniformly contiguous than real cadastre shaped by ridges, roads, and water channels.

Compatibility shim. A thin `synthetic_env_loader.py` reads the generator outputs (`blocks.csv` + `adjacency.json`) and returns the same data structure that `county_env.py` expects from real data; no environment-code changes are required. The synthetic and real evaluation regimes therefore use identical simulator semantics, ensuring that any cross-regime difference is attributable to the landscape, not to the engine.

Restoration-prioritisation env (cross-domain verification)

The cross-domain verification of §6 reuses the entire farmland pipeline (sample, train, plan) by swapping only the env class and the per-unit feature vector. The restoration env is a finite-action gymnasium environment over n_{units} planning cells; the action space is $\{0, 1, \dots, n_{\text{units}} - 1\}$, and “select unit a ” is a one-time monotonic assignment (no swap, no reversibility). The action mask excludes already-selected units and units whose individual restoration cost would exceed the remaining budget. The reward at step t when selecting unit a_t is a fixed linear combination

$$r_t = \sum_k w_k \cdot \phi_k(a_t) + w_{\text{conn}} \cdot \frac{|\{j \in \mathcal{N}(a_t) : j \text{ already selected}\}|}{|\mathcal{N}(a_t)|} - w_{\text{cost}} \cdot \frac{c(a_t)}{B}$$

where ϕ_k are per-unit attribute scores read from a CSV (risk index, water-quality priority, bio-diversity, etc.), w_k are weights specified in a per-case scenario configuration file, $\mathcal{N}(a_t)$ is the queen-contiguity neighbour set, $c(a_t)$ is the unit’s restoration-cost proxy, and B is the cumulative budget. State features $b \in \mathbb{R}^{n_{\text{units}} \times 17}$ are the standardised per-unit attributes (zero-padded to match $K_{\text{block}} = 17$ so the same network architecture as farmland applies); the global state $g \in \mathbb{R}^{12}$ encodes fraction-of-budget-remaining, fraction-of-units-selected, fraction-of-steps-elapsed, and per-component cumulative reward. Snapshot/restore is implemented by capturing the boolean selection mask, cumulative reward components, budget used, and step counter; the env is purely combinatorial and has no expensive-to-restore geometric state, so pairwise dataset generation runs in seconds rather than minutes.

For the Buchanan VA case ($n_{\text{units}} = 562$), per-unit features are derived from the OSMRE e-AMLIS national abandoned-mine-lands inventory [Office of Surface Mining Reclamation and Enforcement, 2019], USGS NHD flowline data [U.S. Geological Survey, 2024b], USGS 3DEP elevation [U.S. Geological Survey, 2024a], and Census TIGER county boundaries [U.S. Census Bureau, 2023], all freely available as public open data; episode length is 50 steps and the cumulative cost cap is set to \$200,000 in notional cost-proxy units. The cost-proxy field is derived as `restoration_cost_proxy` = unit area in hectares (rescaled by 10), which yields a roughly e-AMLIS-comparable ranking among units but should not be interpreted as a real-world dollar reclamation cost; the e-AMLIS unfunded-cost field is sparse and unevenly populated across the inventory, and we therefore use a transparent area-based proxy and report results in proxy units rather than claim a calibrated cost-benefit analysis. For the synthetic case ($n_{\text{units}} = 420$), feature, adjacency, and reward generators are deterministic given a seed and are released alongside the synthetic benchmark of §5; episode length is 60 steps and the cost cap is 45,435 in synthetic units (no real-world correspondence). Both cases share the same MPC configuration as farmland ($H=5$, $K=50$, greedy continuation, scoring=reward) and the same training hyperparameters ($\lambda_{\text{rank}}=5.0$ for contrastive, margin 0.1, three-member ensembles per seed, five seeds for both the contrastive ($\lambda=5$) and the MSE-only ($\lambda=0$) conditions to ensure a fair head-to-head comparison.

The cumulative wall-clock for the full restoration verification (two cases, one prepare each, one sample each, six ensemble trains per case, six MPC plans, one random baseline per case) is approximately 30 minutes on a 14-core M-series Apple Silicon CPU running 3 ensembles in parallel; the cases are released as part of the open reproduction kit (§Code Availability).

Simulator-cost crossover protocol

The simulator-cost analysis of §7 treats the per-step wall cost of `env.step` as an independent variable. To isolate it from confounds (action space size, reward function, optimisation algorithm), we instrument the restoration env’s reward dispatcher with a configurable `time.sleep(delay_s)`

call invoked exactly once per `env.step`. We then run two algorithms at matched 30 s wall budgets across delays $d \in \{0, 1, 10, 50, 100\}$ ms.

Simulated-annealing baseline. We run a permutation-space SA over the candidate-unit action ordering. Each iteration: sample $i, j \in \{0, \dots, 79\}$ uniformly; swap positions i and j in the current ordering; re-roll out the full episode; accept the swap if it improves the cumulative reward, or with probability $\exp(\Delta r / T_t)$ otherwise; geometrically cool T_t at rate 0.995. Initial ordering is the priority-score-greedy ordering (this gives SA a strong warm start). The number of iterations actually completed within the 30 s wall budget is the diagnostic: at delay 0 it reaches $\sim 9,000$, at delay 100 ms it reaches ~ 6 .

Learned-surrogate MPC baseline. We run `mpc.select_action` with random continuation, $H=5$, $K=50$, $\gamma=0.99$, scoring=reward, on each of the seeded RNG paths $\{1, 18, 35, \dots\}$ until the wall budget is exhausted. We report the best cumulative episode reward seen. The per-step ensemble forward pass is ~ 25 ms (one batched ONNX call evaluating all $K=50$ candidates in parallel) and is decoupled from the injected `env.step` delay; only the eventual environment commit step pays the simulator cost.

Farmland baseline measurement. The farmland Bishan datapoint (Figure 3, gold star) is measured on the actual `CountyLevelEnv` without any injected delay. The natural step cost is profiled as 17.9 ± 0.3 ms over a 5-step probe repeated three times (the variance is dominated by per-step *baimu fang* cache hits / misses; in the worst case where the union-find aggregation triggers, single-step cost rises to ~ 60 ms). Under a 180 s wall budget rather than 30 s (chosen because the wall cost per SA iteration on farmland is ~ 1.8 s, so 30 s would only allow ~ 16 iterations and SA would not even reach its initial cooling regime), simulated annealing reaches reward 9.04 at $n_{\text{iter}} = 100$, random restart over 98 episodes reaches 30.85, and contrastive MPC reaches 86.74. The full driver script is released in `farmland_mpc/tests/farmland_baselines.py`.

ArcGIS Pro toolchain

The deployment artefact described in §8 is an ArcGIS Pro Python Toolbox (`LandUseOptimization.P9.pyt`) exposing five tools: (1) *Prepare Data & Blocks* ingests a Third National Land Survey DLTB shapefile and a digital elevation model and emits the block-level state expected by the learned-model code; (2) *Sample Transitions* generates the pairwise dataset $\mathcal{D}_{\text{pw}}$ on the user’s data via the snapshot/restore protocol described above; (3) *Train Contrastive Ensemble* trains a 3-member ensemble at $\lambda_{\text{rank}}=5.0$ and exports each model to ONNX with the user’s N_{blocks} baked in; (4) *MPC Planning* runs the planning loop and writes back an optimised cadastral shapefile readable by any standard GIS workflow; (5) *Check Dependencies* validates the user’s Python environment (ArcGIS Pro 3.6+, Spatial Analyst extension, `gymnasium`, `libpysal`) before any pipeline run is permitted. Tools 1–3 are run once per county on commodity desktop hardware (~ 1 – 2 hours total wall time); Tool 4 is the iterated planning loop (one ~ 7 -minute scenario per call). A 70-second smoke test on a 2-township subset of Neijiang Dongxing District completes the full pipeline as a continuous-integration check.

Two operational caveats are visible in the toolchain. First, the ONNX export bakes in N_{blocks} , so a pre-trained ensemble cannot be reused across counties; Tool 4 carries an explicit pre-flight check that refuses to run if the pre-trained ensemble’s N_{blocks} does not match the prepared dataset. Second, the default projected coordinate reference system (EPSG:32648, UTM zone 48N) covers central and western China; users in eastern, north-eastern, or north-western China must override

the projection in Tool 1 to avoid slope errors. The end-to-end CLI Bishan replication on the package-side prepare achieves a slope improvement of $-1.5439 \pm 0.0409\%$ (1 shipped ensemble, 5 episodes), bracketed by the lab-pipeline matched-DRL figure of $-1.289 \pm 0.079\%$ (5 independently trained ensembles, 1 episode each, §4) and the on-disk GIS verification anchor of -2.001% (§9); the pipeline-to-pipeline gap reflects the corrected geographic-CRS slope step in the package-side prepare (Methods, §9), which exposes truer initial slopes ($\sim 9.62^\circ$) than the lab pipeline’s 8.41° and therefore leaves more reduction headroom. A complete cross-pipeline reconciliation table is reported in Supplementary Section S2.

Evaluation protocol

For the real-data experiments (§4), each configuration is evaluated with 5 deterministic episodes on the real CountyLevelEnv. Reported metrics include mean reward, slope change (%), contiguity change, *baimu fang* count change, and total *baimu fang* area change. For the synthetic benchmark (§5), each method is run for 5 random seeds on each of the 7 presets; per-cell results are aggregated as cross-seed mean $\pm$ sample standard deviation. Cross-seed statistics for the contrastive configurations are computed across 5 independently trained ensembles (different random seeds for both the pairwise dataset and the network initialisation). Tests against fixed reference values use one-sample *t*-tests; superiority comparisons between two methods on independent seeds use Welch’s two-sample *t*-tests; superiority comparisons between two configurations on matched seeds use paired *t*-tests, and equivalence at a stated reward-units margin uses two-one-sided-tests (TOST); ranking-distribution comparisons additionally use Wilcoxon signed-rank as a robustness check. Reported quantities are test type, df, *p*, and Cohen’s *d* (or d_z for paired tests).

Independent GIS-grounded verification of reported gains

Because the headline metrics in §4 are computed inside the same CountyLevelEnv whose learned ensemble drives the MPC scoring loop, a sceptical reviewer can ask whether the reported slope reduction is a real change in the optimised cadastre or an artefact of the learned dynamics agreeing with the planner. To resolve this, we recompute all three primary metrics directly from the optimised DLTB shapefile written to disk by Tool 4, using a 190-line script that depends only on `geopandas` and `libpysal.weights.Queen` and shares no source with the optimisation pipeline (no `torch`, no `ONNX`, no `env` class). The script reads the per-parcel `CHG_FLAG` audit field that Tool 4 emits, reconstructs the post-optimisation land-use vector, joins per-parcel `slope_mean` from the prepared DEM-derived raster on BSM, reprojects to EPSG:32648 for area computation, builds Queen contiguity, and applies the metric definitions exactly as stated in §2 to both the original and the optimised land-use vectors. This verification operates on the package-side prepare (Tool 1 output, 2,640 blocks, 53,004 swappable parcels) and a single canonical episode of the shipped ensemble.

On the 53,086-parcel Bishan deployment dataset, the recomputed deltas (independent path) and the env-reported deltas (planning path) agree to floating-point reduction noise: slope -2.0000% vs. -2.0006% ($|\Delta|=6 \times 10^{-4}$ pp); contiguity $+0.012739$ vs. $+0.012748$ ($|\Delta|=8 \times 10^{-6}$); *baimu fang* area -473.30 ha vs. -473.30 ha ($|\Delta| < 10^{-4}$ ha); and the *baimu fang* count change reproduces exactly at +8 patches. The 428 farm→forest and 428 forest→farm swaps written to the shapefile are also bit-identical to the env’s tally, confirming the farmland-count conservation invariant on disk. The headline gain is therefore a property of the on-disk geometry, not of the learned dynamics model.

We disclose two transparency caveats. First, the lab-pipeline five-ensemble sweep cited in §4 is by construction deterministic at planning time: under fixed ensemble member ordering, top-*K* rollout scoring, and greedy continuation, the seed initialises RNGs that are not reached on the

action-selection path; the cross-seed standard deviation in that comparison comes from the five *independently trained ensembles*, not from re-rolling a single ensemble. To surface a meaningful variance estimator on the package-side single-ensemble protocol used here, we replaced seed jitter with *ensemble member subset ablation*, re-running MPC three times under the three distinct 2-of-3 member subsets. Across these replicates the slope reduction is $-1.998 \pm 0.026\%$ (range $[-2.022, -1.971]$), contiguity change is $+0.0131 \pm 0.0016$, and *baimu fang* area change is -474.5 ± 31.1 ha—sandwich-bracketing the full-ensemble result of -2.001% , $+0.01275$, -473.3 ha to within sample standard deviation on every metric. The full ensemble therefore sits at the median of subset performance rather than benefiting from any single member, and operational variability under the strongest available counter-factual (a different 2-of-3 subset) is bounded at ± 0.026 pp on slope. Second, the verification script keeps 53,086 swappable parcels (those with both a slope value and a farm/forest `ORIG_DLBM`), 82 more than the env’s 53,004; the excess corresponds to parcels whose township code falls outside the configured registry, all of which carry `CHG_FLAG` = 0 and therefore do not affect the delta computation. The verification script is released alongside the toolchain (see *Code availability*) so that any third party with Tool 4’s optimised shapefile and a slope raster can reproduce the headline numbers without invoking the optimisation pipeline.

Direct measurement of ensemble 1-step prediction accuracy

The on-disk verification above only rules out errors in the writer or the metric definitions; it does not bound the contribution of dynamics-model error to the planner’s choices. To address that, we rolled the same MPC policy ($H=5$, $K=50$, $\gamma=0.99$, greedy continuation, reward scoring) for 100 steps on the Bishan deployment dataset and recorded, at every step, both the ensemble’s prediction and the true `env.step` output. The action sequence and per-step metrics reproduce episode 0 of §4 to floating-point identity (slope at steps 10/20/.../100 matches to four decimals), confirming determinism of the deployed pipeline.

Across the 100 in-distribution steps, the ensemble’s reward predictions have MAE 0.818 against an empirical $\sigma(r_{\text{true}}) = 1.40$, with predictions confined to a narrow band of width 0.24 (range $[+0.43, +0.67]$) while the true reward varies in $[-3.04, +6.36]$. The ensemble is therefore strongly mis-calibrated as a regressor, scoring barely better than a constant predictor (naive-mean baseline MAE 0.83). Despite this, the Spearman rank correlation between predicted and true reward is $+0.22$, and this is what MPC’s top- K selection consumes; the trajectory match to §4 confirms that ranking signal is sufficient at $H=5$ horizon. Next-state predictions for the slope-improvement channel (*gf*[4], used directly when `scoring="slope"`) carry MAE 2.4×10^{-3} , smaller than the channel’s own per-step standard deviation (5.2×10^{-3}). These three numbers together are a direct empirical instantiation of the ranking-versus-regression decomposition we developed in §3: the contrastive ensemble has not learned to predict reward magnitude, but it has learned to rank candidate actions, and the planner’s $H=5$ top- K mechanism converts that weak ranking signal into a deterministic, physically-correct trajectory.

A third transparency caveat follows from this measurement: ensemble disagreement is not informative as an uncertainty signal under the current bootstrap diversity. The mean of r_{std} is 0.027 across the 100 steps, the Pearson correlation between r_{std} and the per-step absolute reward error is $+0.08$, and the maximum disagreement seen (0.075) is two orders of magnitude smaller than the maximum absolute error (5.07). Members under-fit the reward distribution similarly to one another, so disagreement does not flag the cases where they jointly are wrong. Practitioners who want disagreement-driven candidate filtering should train Tool 3 with stronger diversity (different feature subsets or training data slices, not just initialisation seed). The default `scoring="reward"` pathway in Tool 4 remains safe because top- K ranking, not absolute reward, drives planning;

`scoring="slope"` is an alternative that bypasses the reward head entirely and uses the better-calibrated `gf[4]` channel.

Replacing the ensemble with the true environment

As a final ablation we re-ran the same MPC procedure ($H=5$, $K=50$, $\gamma=0.99$, single rollout per candidate, seed 0) but used the actual `env.step` in place of every ensemble call. Two implementation concessions are required to make this tractable: stage 1 ranks a random subset of 200 valid actions per step instead of all $\sim 2,000$ (the ensemble’s `batch_predict` scores every action at near-constant marginal cost; a true-env stage 1 must run incremental simulation per action), and stage 2 uses random continuation instead of greedy. Mutable env state (`land_use`, `swapped`, `farmland_nbr_count`, accumulators, *baimu* cache) is snapshotted before each trial step and restored after; the *baimu fang* union-find is frozen inside scoring trials with the outer commit step computing it correctly. Total wall time was 358s on the same hardware that ran the 1,448s ensemble baseline.

The resulting trajectory finishes at slope -1.819% , contiguity $+0.0087$, and *baimu fang* area change -565 ha, all three strictly worse than the package-side ensemble pipeline’s -2.001% , $+0.0128$, -473 ha. Per-step inspection shows true-env MPC starting behind (the 200-action stage 1 misses the strongest moves the ensemble identifies via full-action coverage), narrowing the gap as the trajectory matures, and finishing 0.18 pp short of the ensemble’s slope. The reason is structural rather than statistical: ensemble `batch_predict` grants the planner the depth and breadth that incremental simulation cannot afford within the same compute budget. Putting matched effort into true-env MPC—full enumeration in stage 1 and greedy continuation in stage 2—would cost approximately five days of wall time per episode for a payoff of at most a fraction of a percentage point of slope. The headline -2.00% result is therefore on the optimistic side of the planner’s accessible frontier under matched compute, and the ensemble’s contribution to the deployment is structural (it grants the planner depth and breadth) rather than illusory.

Data availability

The raw farmland cadastral records for Bishan District (Chongqing) and Neijiang Dongxing District (Sichuan) are derived from the Third National Land Survey of China and cannot be publicly redistributed under existing data-governance restrictions. Researchers requiring access to the raw cadastral records should approach the Chongqing and Sichuan provincial natural-resources departments under the same procedure used for any access to Third Land Survey data; we have no role in granting access. To support reproducibility and verification independent of restricted access, all derived data products required to reproduce the headline experiments are publicly available now (not gated on publication) at the project repository (§Code Availability) under MIT licence:

(a) the full synthetic farmland benchmark of seven landscape presets with deterministic generators (Section 5) under CC-BY 4.0; this benchmark alone is end-to-end sufficient to reproduce the contrastive-MPC pipeline’s qualitative findings and to benchmark future methods without any access to restricted data; (b) aggregated block-level features (the 17-dimensional vectors for all 2,600 Bishan and 3,711 Neijiang blocks in the initial state) and the anonymised pairwise dataset $\mathcal{D}_{pw}$ for both farmland counties, sufficient to reproduce transition-model training, contrastive training, and MPC evaluation end-to-end without raw parcel records; (c) the released contrastive ensembles for both Bishan and Neijiang (5×3 trained members per case, plus the bishan λ -ablation set used in §3); (d) random seeds, hyperparameters, and de-identified training logs for all reported farmland experiments; (e) the complete restoration cases of §6: *Buchanan County, Virginia, USA* (planning-unit attributes, adjacency, OSMRE e-AMLIS extracts, USGS NHD flowlines clipped to the county,

USGS 3DEP slope, and Census TIGER county boundary, all already public per their respective US-government open-data licences) and the deterministic *synthetic* mine-restoration generator with all seven weight-sweep Pareto configurations.

The synthetic farmland benchmark (a) and the public-data Buchanan restoration case (e) jointly form a fully open reproduction track that requires no restricted-data access; together with the released ensembles (c), an external reviewer or future user can reproduce all qualitative findings of this paper end-to-end on commodity desktop hardware. The cadastral derivatives (b)–(d) extend that reproduction to the actual real-county Bishan and Neijiang results without requiring the raw cadastral parcels themselves, under the same data-governance constraints that apply to many Earth science datasets covering politically sensitive regions.

Code availability

All code for the contrastive training, MPC planner, synthetic generators, ArcGIS Pro toolchain, the non-commercial CLI workflow, the operations-research baselines (§6 Table 2: greedy, simulated annealing, NSGA-II, CBC-MILP), the planner-relevant ranking metrics (NDCG, top- K regret), and the Pareto-sweep driver is released as a public repository on GitHub at the time of submission (<https://github.com/zhouning/arcgis-farmland-mpc>), under an MIT licence, with step-by-step reproduction instructions in `docs/REPRODUCE.md` and a smoke test that completes a full Tool 1–4 cycle in ~ 70 seconds on a small subset.

Author contributions

N.Z. conceived the study, designed and implemented the methods, conducted all experiments, and wrote the manuscript. X.J. supervised the research and revised the manuscript.

Competing interests

The authors declare no competing interests.

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References

- Jeroen C. J. H. Aerts and Gerard B. M. Heuvelink. Using simulated annealing for resource allocation. *International Journal of Geographical Information Science*, 16(6):571–587, 2002.
- Christopher Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. Learning to rank using gradient descent. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 89–96, 2005. doi: 10.1145/1102351.1102363. URL <https://www.microsoft.com/en-us/research/publication/learning-to-rank-using-gradient-descent/>. Seminal pairwise learning-to-rank work (RankNet; cross-entropy over pairwise preferences).
- Christopher J. C. Burges. From RankNet to LambdaRank to LambdaMART: An overview. Technical report, Microsoft Research Technical Report MSR-TR-2010-82, 2010.
- Kai Cao, Bo Huang, Shaowen Wang, and Hui Lin. A boundary-based generic genetic algorithm for sustainable land-use planning. *Computers, Environment and Urban Systems*, 36(3):257–269, 2012. doi: 10.1016/j.compenvurbsys.2011.08.001. CEUS, genetic-algorithm baseline for sustainable land-use planning; representative pre-DRL method on closely related task.
- Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL <https://papers.nips.cc/paper/2017/hash/d5e2c0adad503c91f91df240d0cd4e49-Abstract.html>.
- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL <https://papers.nips.cc/paper/2018/hash/3de568f8597b94bda53149c7d7f5958c-Abstract.html>.
- Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. A. M. T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002.
- Julian Equihua et al. Connectivity conservation planning through deep reinforcement learning. *Methods in Ecology and Evolution*, 2024. DRL applied to connectivity-aware conservation; related precedent for combinatorial spatial planning with ecological objectives.
- European Space Agency. Copernicus DEM – global and European digital elevation model (COP-DEM), 2021. Product handbook GLO-30, version 1.1; available via <https://copernicus-dem-30m.s3.amazonaws.com/>.
- Jonathan A. Foley, Navin Ramankutty, Kate A. Brauman, Emily S. Cassidy, James S. Gerber, et al. Solutions for a cultivated planet. *Nature*, 478(7369):337–342, 2011.
- Carlos E. García, David M. Prett, and Manfred Morari. Model predictive control: Theory and practice—a survey. *Automatica*, 25(3):335–348, 1989.
- David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL <https://papers.nips.cc/paper/2018/hash/2de5d16682c3c35007e4e92982f1a2ba-Abstract.html>.

- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *Nature*, 640:647–653, 2025. doi: 10.1038/s41586-025-08744-2. URL <https://www.nature.com/articles/s41586-025-08744-2>.
- Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=0xh5CstDJU>. Spotlight.
- Ralf Herbrich, Thore Graepel, and Klaus Obermayer. Large margin rank boundaries for ordinal regression. In *Advances in Large Margin Classifiers*, pages 115–132. MIT Press, 2000. Original large-margin pairwise ranking formulation for ordinal regression.
- Jikun Huang, Wei Wei, Qi Cui, and Wei Xie. The prospects for china’s food security and imports. *China Agricultural Economic Review*, 2018.
- Shengyi Huang and Santiago Ontañón. A closer look at invalid action masking in policy gradient algorithms. In *The International FLAIRS Conference Proceedings*, 2022.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://papers.nips.cc/paper/2019/hash/5faf461eff3099671ad63c6f3f094f7f-Abstract.html>.
- Krzysztof Janowicz, Song Gao, Grant McKenzie, Yingjie Hu, and Budhendra Bhaduri. GeoAI: Spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. *International Journal of Geographical Information Science*, 34(4):625–636, 2020. doi: 10.1080/13658816.2019.1684500. GeoAI overview; situates AI methods in geographic knowledge discovery.
- Thorsten Joachims. Optimizing search engines using clickthrough data. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 133–142, 2002. doi: 10.1145/775047.775067. Ranking SVM; large-margin pairwise ranking in its hinge formulation—the direct ancestor of our margin ranking loss.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL <https://papers.nips.cc/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html>.
- Y. Liu and Y. Li. Farmland fragmentation and its impact on agricultural productivity: Evidence and implications for land consolidation policy. *Land Use Policy*, 2023. Domain motivation for land consolidation; fragmentation–productivity link.
- Yansui Liu, Fang Fang, and Yuheng Li. Land consolidation projects in rural china: Origins, progress, and outcomes. *Land Use Policy*, 40:3–14, 2014.
- Hualou Long. Land use policy in china: Introduction. *Land Use Policy*, 40:1–2, 2014.
- S. Niu et al. Quantifying farmland fragmentation: Indicator system and spatial–temporal patterns. *Land*, 2023. [v7-PLACEHOLDER] Verify full author list, volume, issue, and DOI before submission. Used in v7 §1 to support “per-hectare labour cost, mechanisation barriers, climate-driven yield-shock vulnerability” framing. Aligned with the FFI implementation in the author’s own `FFI.py`.

- Office of Surface Mining Reclamation and Enforcement. Abandoned mine land inventory system (e-AMLIS), 2019. Public ArcGIS REST service; inventory snapshot September 2019; <https://amlis.osmre.gov/>.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022. URL <https://papers.nips.cc/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract-Conference.html>.
- Markus Reichstein, Gustau Camps-Valls, Bjorn Stevens, Martin Jung, Joachim Denzler, Nuno Carvalhais, and Prabhat. Deep learning and process understanding for data-driven Earth system science. *Nature*, 566(7743):195–204, 2019.
- David Rolnick, Priya L. Donti, Lynn H. Kaack, Kelly Kochanski, Alexandre Lacoste, et al. Tackling climate change with machine learning. *ACM Computing Surveys*, 55(2):1–96, 2022.
- Rosalaura Romeo, Sara Manuelli, Mariella Geringer, and Valeria Barchiesi. Mountain agriculture: opportunities for harnessing zero hunger in Asia. FAO report, 2020. [v7-PLACEHOLDER] Verify exact title, authorship, year, and pagination before submission. Used in v7 §1 and Abstract framing for “roughly a fifth of global agricultural production from mountainous landscapes”. If the precise figure cannot be sourced from this report, drop the quantification or replace with a Mountain Partnership statistical reference.
- Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 15, pages 627–635. PMLR, 2011. URL <https://proceedings.mlr.press/v15/ross11a.html>.
- Mark D. A. Rounsevell, Bas Pedroli, Karl-Heinz Erb, Marc Gramberger, Anne Gravsholt Busck, et al. Challenges for land system science. *Land Use Policy*, 29(4):899–910, 2012.
- Reuven Y. Rubinstein. The cross-entropy method for combinatorial and continuous optimization. *Methodology and Computing in Applied Probability*, 1(2):127–190, 1999.
- Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, Thore Graepel, Timothy Lillicrap, and David Silver. Mastering Atari, Go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, 2020. doi: 10.1038/s41586-020-03051-4. URL <https://www.nature.com/articles/s41586-020-03051-4>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- David Silver, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, et al. Mastering the game of go without human knowledge. *Nature*, 550(7676):354–359, 2017.
- Marco Springmann, Michael Clark, Daniel Mason-D’Croz, Keith Wiebe, Benjamin Leon Bodirsky, et al. Options for keeping the food system within environmental limits. *Nature*, 562(7728): 519–525, 2018.
- Theodor J. Stewart, Ron Janssen, and Marjan van Herwijnen. A multiobjective genetic programming approach to land use planning. *Computers & Operations Research*, 31(14):2293–2313, 2004.

- Devis Tuia, Konrad Schindler, Begum Demir, Xiao Xiang Zhu, Mrinalini Kochupillai, et al. Artificial intelligence to advance Earth observation: A review of models, recent trends, and pathways forward. *IEEE Geoscience and Remote Sensing Magazine*, 2024.
- U.S. Census Bureau. TIGER/line cartographic boundary files, 2023. 500K-resolution county boundaries; <https://www.census.gov/geographies/mapping-files/>.
- U.S. Geological Survey. 3d elevation program (3DEP), 2024a. Elevation imagery and derived slope rasters; <https://www.usgs.gov/3d-elevation-program>.
- U.S. Geological Survey. National hydrography dataset (NHD), 2024b. Hydrography MapServer; <https://hydro.nationalmap.gov/arcgis/rest/services/nhd/>.
- Grady Williams, Paul Drews, Brian Goldfain, James M. Rehg, and Evangelos A. Theodorou. Model predictive path integral control: From theory to parallel computation. *Journal of Guidance, Control, and Dynamics*, 40(2):344–357, 2017.
- Y. Zheng, Y. Lin, L. Zhao, et al. Reinforcement learning for community spatial planning. *Nature Computational Science*, 2023. Deep-RL applied to urban community spatial planning; reference for DRL in structured discrete spatial design.